Fixture note

Fixture note: section 1

The solar system formed about 4.6 billion years ago from the collapse of a giant molecular cloud. Stellar black holes form when massive stars exhaust their nuclear fuel and collapse under their own weight. Comets are icy bodies that grow bright tails when their orbits carry them close to the Sun. The asteroid belt between Mars and Jupiter holds the leftover building blocks of the early solar system. Renewable energy comes from sources that are replenished naturally, such as sunlight, wind and flowing water. Hydropower is the largest source of renewable electricity, though new dams raise ecological questions.

The Milky Way contains hundreds of billions of stars, most of them orbiting a central supermassive black hole. Gravitational waves from merging black holes were first detected in 2015, opening a new way to observe the universe. Solar panels convert sunlight directly into electricity, and their cost has fallen sharply over the past decade. Energy storage, from lithium batteries to pumped hydro, smooths out the variable output of wind and solar. Biomass can replace fossil fuels in heating and transport, but sustainable sourcing is essential. Good results come from patient preparation and from measuring ingredients and timings carefully.

Wind turbines turn the kinetic energy of moving air into electrical power, onshore and offshore. Geothermal plants use heat from the Earth, which makes them steady sources of baseload power. Capacity factor measures how much of its maximum output a power plant actually delivers over a year. Beginners often skip the basics, then spend more time repairing mistakes than they would have spent learning. A small amount of regular practice beats an occasional long session. Les planètes tournent autour du Soleil, et les quatre planètes intérieures sont petites et rocheuses.

Heat pumps move heat rather than creating it, delivering three or more units of heat per unit of electricity. Local conditions such as weather, soil or water quality change what works from place to place. Tools do not need to be expensive, but they should be well maintained and the right size for the job. Un trou noir est une région de l’espace où la gravité est si forte que même la lumière ne peut s’en échapper. Planets orbit the Sun in nearly circular paths, and the four inner planets are small and rocky. Telescopes collect faint light, letting astronomers measure the distances, motions and chemistry of distant objects.

Le système solaire s’est formé il y a environ 4,6 milliards d’années à partir d’un nuage de gaz et de poussière. Les astronomes utilisent des télescopes pour mesurer la distance et la composition des étoiles lointaines. A black hole is a region of spacetime where gravity is so strong that nothing, not even light, can escape. The giant planets Jupiter and Saturn are mostly hydrogen and helium, with deep atmospheres and many moons. Astronomers detect exoplanets by watching for tiny dips in starlight as a planet crosses in front of its star. Orbital periods grow longer with distance: Mercury takes 88 days, while Neptune takes about 165 years.